Sustainable Urban Logistics Optimization

Capstone Project Guide

This document contains the complete requirements and evaluation criteria for your course project.

Contents

1 Project Overview

2 Business Objectives

3 Available Data

4 Implementation Guidance

5 Specific Client Case Studies

6 Deliverables

7 Evaluation Framework

- **7.1 Project Quality (60%)**
- **7.2 Presentation Quality (35%)**
- **7.3 Individual Contribution (5%)**

8 Technical Requirements

9 Resources

10 Tips for Success

11 Important Notes

12 Team Formation Requirements

- **12.1 Team Formation Process**

Appendix: CAPSTONE CHECKLIST. HOW TO GET A 4 (EXCELLENT)

1 Project Overview

Your team has been hired as consultants by **GreenCity Logistics**, a company that specializes in providing sustainable last-mile delivery solutions for urban areas. The city government has initiated a program to reduce traffic congestion and carbon emissions while maintaining efficient delivery services. GreenCity Logistics wants to establish a network of micro-fulfillment centers that will optimize delivery operations while meeting environmental goals. Among their major clients, several have specific operational requirements that add complexity to the optimization problem. For instance, Client C103 wants to limit vehicle routes to at most 6 different stops per trip for quality control, requires electric vehicles in certain districts, and will only use the service if delivery times are guaranteed within 2-hour windows. Client C101, a pharmaceutical distributor, requires refrigerated transportation in specific temperature-controlled increments and wants to minimize the number of different fulfillment centers handling their products.

This project will apply prescriptive analytics techniques from our course to solve a real-world logistics optimization problem.

Key Success Factors:

- Understand the interrelationship between facility location decisions, delivery routing, and environmental impact
- Consider this as a multi-echelon supply chain problem where micro-fulfillment centers serve as an intermediate layer between warehouses and customers
- Focus on balancing economic, environmental, and service level objectives
- Apply multiple optimization techniques (linear, network, integer, Non-linear, and multi-objective) in an integrated solution

2 Business Objectives

Your solution should address the following business objectives:

- Minimize total system cost (facility setup, operations, transportation)
- Minimize environmental impact (carbon emissions, noise, congestion)
- Meet service level requirements (delivery time windows, capacity constraints)
- Create a sustainable and scalable urban logistics network

Multi-Objective Optimization Approaches:

- Weighted Sum Method: Combine objectives using weights based on priorities
- ϵ-Constraint Method: Optimize one objective while constraining others
- Goal Programming: Set target values for each objective and minimize deviations
- Hierarchical Optimization: Solve objectives sequentially based on importance. These objectives naturally conflict with each other - cost-effective solutions may have higher emissions, while environmentally optimal solutions may be more expensive.

3 Available Data

The following datasets will be provided for your analysis:

Project Datasets:

- **location data.csv: Potential Micro-Fulfillment Center Locations**
  - Information about 25 potential locations for establishing fulfillment centers
  - Key fields: location id, district, building type, available area sqm, max capacity packages, setup cost, monthly lease, utility cost, suitability score, zoning restriction
- **demand data.csv: Customer Demand Zones**
  - Information about 30 demand zones
  - Key fields: zone id, district, population density, daily demand packages, priority delivery percent, time window start, time window end, delivery difficulty
- **vehicle data.csv: Vehicle Fleet Options**
  - Details about 10 different vehicle types
  - Key fields: vehicle type, capacity packages, purchase cost, lifespan years, maintenance cost annual, fuel type, fuel efficiency, co2 emissions kg per km, operating cost per km, max range km
- **distance matrix.csv: Travel Distance Matrix**
  - Distances and travel times between facilities and demand zones
  - Key fields: origin, destination, distance km, travel time mins, congestion factor, emission factor, toll cost, time reliability
- **emission constraints.csv: Environmental Constraints**
  - Environmental restrictions for each zone
  - Key fields: zone id, emission cap kg per day, noise restriction level, low emission zone, congestion charge, time restrictions, sustainability incentive

Data Preprocessing Best Practices:

- **Spatial Analysis**: Use clustering to identify high-demand zones
- **Data Normalization**: Ensure distance and time metrics have comparable scales
- **Parameter Conversion**: Transform environmental constraints into quantifiable parameters
- **Data Integration**: Create a consolidated dataset linking all parameters
- **Temporal Considerations**: Account for time-of-day variations in travel times
- **Environmental Metrics**: Calculate emission factors for each vehicle type/route
- **Pro Tip**: Create a data dictionary documenting all variables, units, and relationships to ensure consistency throughout your analysis.

4 Implementation Guidance

1. **Data Analysis**:
    a. Calculate demand patterns and geographic distribution
    b. Analyze correlations between distance, time, and costs
    c. Create heat maps for demand and environmental constraints
    d. Consider time-varying travel patterns and congestion effects
2. **Linear Programming Implementation**:
    a. Start with a basic facility location and allocation model
    b. Implement budget and capacity constraints
    c. Formulate continuous flow variables for initial optimization
    d. Optimize for a single objective initially (e.g., total cost minimization)
    e. Analyze the feasible region and basic optimal solutions
3. **Sensitivity Analysis:**
    a. Determine shadow prices for binding constraints
    b. Calculate allowable increases and decreases for key parameters
    c. Perform parametric analysis on critical input values
    d. Create sensitivity reports for management decision-making
    e. Identify which constraints and parameters most affect the solution
4. **Network Modeling:**
    a. Represent the logistics network as nodes and arcs
    b. Model flow balance constraints at each node
    c. Visualize network structure and bottlenecks
    d. Determine optimal routes
    e. Identify critical paths and network resilience

5. **Integer and Mixed Integer Programming:**
    a. Implement binary variables for facility location decisions
    b. Model vehicle selection and route assignment with integer variables
    c. Use binary variables for ”yes/no” service decisions
    d. Implement cardinality constraints to limit stops per route
    e. Model fixed costs and threshold requirements
6. **Multiple Objective Optimization:**
    a. Implement goal programming to balance competing objectives
    b. Model preference structures using weights or preemptive priorities
    c. Balance cost, emissions, and service level objectives
    d. Implement target constraints with deviation variables
    e. Analyze trade-offs between competing objectives
    f. Create efficient solutions for different stakeholder preferences
7. **Nonlinear Programming Modeling:**
    a. Model congestion effects with nonlinear travel time functions
    b. Implement emissions models that vary with speed and load
    c. Consider volume-dependent operational costs
    d. Account for economies of scale in facility sizing
    e. Analyze the impact of nonlinear constraints on solution quality

Mathematical Formulation Tips:

Linear Programming Base Model:

$$\min_{x,y} \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} + \sum_{i \in I} f_i y_i$$

Subject To:

$$\sum_{i \in I} x_{ij} = d_j, \quad \forall j \in J$$

$$\sum_{j \in J} x_{ij} \leq M_i y_i, \quad \forall i \in I$$

$$x_{ij} \geq 0, \forall i \in \mathrm{I}, \mathrm{j} \in \mathrm{J}$$

$$y_i \in [0,1], \forall i \in \mathrm{I}$$

Where x_{ij} is the flow from facility i to demand zone j, y_i is the capacity utilization of facility i, c_{ij} is the transportation cost, f_i is the facility fixed cost, d_j is the demand at zone j, M_i is the capacity of facility i.

Network Flow Formulation:

$$\min_{x} \sum_{(i,j)\in A} c_{ij}x_{ij}$$

Subject To:

$$\sum_{j:(i,j)\in A} x_{ij} - \sum_{j:(j,i)\in A} x_{ji} = b_i, \quad \forall i \in N$$

$$l_{ij} \leq x_{ij} \leq u_{ij}, \forall (i,j) \in \mathrm{A}$$

Where x_{ij} represents the flow on arc (i,j), to demand zone j, c_{ij} is the cost per unit flow, b_i is the supply/demand at node i, l_{ij}, u_{ij} are lower/upper bounds on flow.

Integer Programming Constraints:

$$x_{ij} \leq Mz_{ij}$$

$$\sum_{j\in J} z_{ij} \leq K_i y_i$$

$$\sum_{i\in I} y_i \leq P$$

$$y_i, z_{ij} \in \{0,1\}, \forall i \in \mathrm{I}, \mathrm{j} \in \mathrm{J}$$

Where y_i indicates whether facility i is opened, z_{ij} indicates whether zone j is served by facility i, K_i is the maximum number of zones that facility i can serve, and P is the maximum number of facilities.

Vehicle Assignment Constraints:

$$\sum_{v\in V} w_{ijv} = z_{ij}$$

$$\sum_{i\in I}\sum_{j\in J} q_j w_{ijv} \leq Q_v$$

$$w_{ijv} \in \{0,1\}, \forall i \in \mathrm{I}, \mathrm{j} \in \mathrm{J}$$

Where w_{ijv} indicates whether vehicle type v is used on route from i to j, q_j is the demand quantity at zone j, and Q_v is the capacity of vehicle type v.

Multiple Objective Formulation (Goal Programming):

$$Min\ \alpha_1(d_1^+ + d_1^-) + \alpha_2(d_2^+ + d_2^-) + \alpha_3(d_3^+ + d_3^-)$$

Subject to:

$$[Cost\ Function] + d_1^+ - d_1^- = g_1$$

$$[Emission\ Function] + d_2^+ - d_2^- = g_2$$

$$[Service\ Level\ Function] + d_3^+ - d_3^- = g_3$$

$$[Other\ Constraints]$$

$$d_1^+, d_1^-, d_2^+, d_2^-, d_3^+, d_3^- \geq 0$$

Where g_1, g_2, g_3 are target values for cost, emissions, and service level, $\alpha_1, \alpha_2, \alpha_3$ are the weights of the target values, and d_i^+, d_i^- are positive and negative deviations from these targets.

Nonlinear Travel Time Function:

$$t_{ij}(x) = t_{ij}^0 \left(1 + \alpha \left(\frac{x}{C_{ij}}\right)^{\beta}\right)$$

Where $t_{ij}(x)$ is the travel time with flow x, t_{ij}^0 is the free-flow travel time, C_{ij} is the capacity of the link, and α, β are parameters.

5 Specific Client Case Studies

You may consider the following specific client cases:

Client Case Study 1: E-Commerce Provider (C103)

– Limit vehicle routes to at most 6 different stops per trip

– Require electric vehicles in certain districts (low emission zones)

– Service must guarantee delivery within 2-hour time windows

– Either serve a zone completely or not at all

Client Case Study 2: Pharmaceutical Distributor (C101)

– Require refrigerated vehicles in specific temperature increments

– Minimize the number of different fulfillment centers handling their products

– Demand strict adherence to time windows for temperature-sensitive products

– Require redundancy in service capability for critical deliveries

Client Case Study 3: Grocery Chain (C108)

– Batch deliveries in specific time blocks

– Manage both ambient and refrigerated products simultaneously

– Require certain minimum volumes to justify service to a zone

– Place caps on emissions for ”green delivery” marketing program

Client Case Study 4: Electronics Retailer (C115)

– Prioritize high-value, time-sensitive deliveries

– Require specialized handling for fragile items

– Limit deliveries to either morning or afternoon blocks

– Ensure each zone is served by at least two fulfillment centers for resilience

6. Deliverables

6.1 Required Deliverables

Each team must submit the following:

1. Technical Report

A professional document that clearly explains your decision system, not just your model.

Required content:

A. Problem Definition

- Clear statement of the decision problem
- Objectives (cost, emissions, service level)
- Scope and assumptions

B. Data Understanding and Preparation

- Description of all datasets used
- Data cleaning and integration steps
- Key derived parameters (costs, emissions, travel feasibility)

- Data limitations (e.g., reduced distance matrix)

C. Mathematical Modeling

You must include:

- Linear Programming (LP) formulation
- Network flow structure
- Mixed Integer Programming (MIP) formulation
- Multi-objective formulation (one method required)

Clearly define:

- Sets
- Parameters
- Decision variables
- Objective function(s)
- Constraints

D. Operational Realism (REQUIRED FOR EXCELLENCE)

You must include at least one of the following:

Option 1 (Recommended):

- Full implementation of one client case study (e.g., C103, C101, C108, C115)

Option 2:

- A self-defined set of realistic operational constraints beyond the base model

For added variables and constraints:

- Explain the business meaning
- Show mathematical formulation
- Justify assumptions if data is not directly available

E. Solution and Results

- Optimal solution(s)
- Facility locations selected
- Demand allocation
- Vehicle usage

F. Sensitivity and Scenario Analysis

You must analyze:

- At least 2 key parameters
- Impact on solution and decisions

G. Managerial Insights (CRITICAL)

- Interpret results in business terms
- Explain trade-offs (cost vs emissions vs service)
- Provide actionable recommendations

H. Limitations and Extensions

- Data limitations
- Modeling simplifications
- Proposed improvements

6.2 Executive Presentation

Video Submission Requirement:
Each team must **create a 10-minute video presentation** of their Executive Presentation. The video should clearly communicate the problem, modeling approach, key results, trade-offs, and business recommendations. This video must be **uploaded to CAMINO** and is **due June 1 at 23:59.** During the last two class sessions (June 2 and 4), each team will play their video in class, followed by a 5-minute live Q&A session with the class.

6.3 Implementation Package

- Code (Colab Notebook, gurobipy, Python,)
- Optimization models
- Data processing scripts
- Visualizations

Requirements:

- Well-structured
- Documented
- Reproducible

6. 4 User Guide

Must include:

- How to run the model
- Input definitions
- Output interpretation
- Limitations and assumptions

7 Evaluation Framework

Evaluation Distribution

Your project will be evaluated according to the following distribution:

Component	Weight
Project Quality	60%
Presentation	35%
Individual Contribution	5%

7.1 Project Quality Criteria (60%)

Your project will be evaluated on these criteria, each scored from 1 (Poor) to 4 (Excellent).

Project Quality evaluates:

- Technical correctness
- Depth of modeling
- Realism of the solution
- Quality of insights

Project Quality Criteria

Each project is evaluated on the following dimensions:

1. Project Completion

- Coverage of all required components:
- LP model
- Network model
- MIP formulation
- Multi-objective optimization

2. Technical Correctness

- Correct formulation of:
    - Objective function(s)
    - Constraints

    - Decision variables
- Logical consistency
- Feasible and valid solutions

Note: Projects without MIP components cannot score above 2 in this criterion.

3. Operational Realism

Evaluates the ability to translate real-world constraints into mathematical models.

High-quality work includes:

- Client-specific constraints (e.g., C103) OR
- Well-justified real-world extensions

Assessment focuses on:

- Correctness of modeling
- Clarity of interpretation
- Justification of assumptions

4. Analysis Quality

- Depth of analysis
- Use of sensitivity/scenario analysis
- Identification of key drivers
- Clear explanation of results

5. Managerial Insight and Decision Support

- Ability to translate results into implementable decisions
- Clarity of recommendations
- Understanding of trade-offs
- Business relevance

Rubric

Score	Description
4 – Excellent	Complete, correct, and well-structured project. Includes **realistic operational constraints (e.g., client case study)**. Demonstrates deep analysis, strong trade-off understanding, and clear, actionable business insights.

Score	Description
3 – Good	Technically correct and complete core model (LP, MIP, multi-objective). Good analysis and results. **Limited or no operational realism beyond basic constraints.**
2 – Satisfactory	Partial implementation. Missing key components or weak modeling. Limited analysis and insight.
1 – Poor	Incomplete, incorrect, or poorly structured work. Major errors in modeling and analysis.

7.2 Presentation Quality Criteria (35%)

Your presentation will be evaluated on these criteria, each scored from 1 (Poor) to 4

(Excellent):

- **Content Delivery**: Clarity, organization, professionalism, and ability to communicate the decision problem, modeling approach, and key insights
- **Visual Aids**: Quality and effectiveness of visual materials, including clear presentation of model outputs, trade-offs, and decisions
- **Q&A Handling**: Ability to address questions accurately, demonstrating understanding of modeling choices, assumptions, and business implications
- **Time Management**: Effective use of allocated presentation time, with appropriate balance between technical explanation and managerial insights
- **Group Coordination**: Seamless transitions and team collaboration, with consistent understanding of both technical and business aspects of the project

High-quality presentations should clearly demonstrate:

- The business context of the problem
- The impact of operational constraints (e.g., client case studies or equivalent)
- The trade-offs between objectives (cost, emissions, service level)
- Actionable recommendations based on the model results

Presentation Tips

- Practice your presentation multiple times as a group
- Ensure smooth transitions between speakers
- Prepare concise explanations of technical concepts

- Anticipate potential questions and prepare answers
- Connect your analysis to business objectives

Table 2: Presentation Quality Rubric

Score	Description
4 - Excellent	Polished delivery with clear articulation, highly professional and engaging presentation. Clearly communicates **decision problem, modeling approach, operational constraints, and business insights**, with effective time use and well-prepared Q&A responses.
3 - Good	Logical flow with smooth presentation and good clarity. Visual aids used effectively to make points. Demonstrates understanding of the model and results, but **limited emphasis on operational realism or business insights**. Coordinated group work.
2 - Satisfactory	Clear but basic presentation, adequate visual aids. Focuses primarily on technical aspects with **limited interpretation of results or decision impact**. Generally stayed on topic with minor coordination issues.
1 - Poor	Disorganized, hard to follow, poor visual aids. Fails to clearly explain the problem, model, or results. Limited connection to business context. Ran over time with poor coordination between speakers.

7.3 Individual Contribution (5%)

Individual Contribution Assessment

You will be evaluated individually on these criteria, each scored from 1 (Poor) to 4

(Excellent):

- **Task Completion**: Fulfilling your assigned responsibilities, including contributions to model development, analysis, or presentation preparation
- **Quality of Work**: Meeting or exceeding quality standards in your contributions, including technical correctness, clarity, and relevance to the decision problem
- **Communication**: Regular updates with team members and ability to clearly explain your work, including modeling choices, assumptions, and results
- **Collaboration**: Active contribution to group efforts, including participation in both technical development and preparation of business insights

- **Reliability**: Meeting deadlines and commitments, and supporting the team in delivering a complete and well-integrated project

Assessment will be based on:

- Individual Q&A during the presentation
- **Peer evaluation from team members**

Rubric

Score	Description
4 – Excellent	Consistently fulfills responsibilities with high-quality contributions. Demonstrates strong understanding of both **technical modeling and business implications**. Actively contributes to **analysis, operational realism, and insights**, and communicates ideas clearly during team work and Q&A. Highly reliable and collaborative.
3 – Good	Completes assigned tasks with good quality. Demonstrates solid understanding of the model and contributes to team efforts. Participates in communication and presentation, but with **limited involvement in deeper analysis or business insights**. Reliable and cooperative.
2 – Satisfactory	Completes some assigned tasks but with uneven quality. Limited contribution to modeling, analysis, or presentation. Communication and collaboration are basic. Shows **partial understanding of the project and its implications**.
1 – Poor	Fails to meet responsibilities or contributes minimally. Work is incomplete or incorrect. Limited participation in team activities and presentation. Poor communication and reliability.

8 Technical Requirements

Implementation Requirements:

- You will use Python in Google Colab Notebooks, with gurobipy as the primary optimization tool
- Document all assumptions clearly
- Include references to academic literature or industry practices where appropriate

- Ensure that your Google Colab Notebook is well-organized and well-documented, and that your mathematical models are clearly explained
- Test your solutions with different client scenarios (or equivalent operational constraints) to demonstrate flexibility

Software Selection Considerations:

Google Colab Notebooks (Required Environment):

- Cloud-based, no local installation required
- Enables reproducibility and easy sharing
- Supports integration of code, results, and explanations in a single document

Python (Required Language):

- Flexible and powerful for optimization modeling
- Supports data processing, modeling, and visualization
- Enables scalable and structured implementations

gurobipy (Required Optimization Library):

- Industry-standard solver for LP, MIP, and advanced optimization models
- Efficient for large-scale and complex decision problems
- Supports multi-objective optimization and advanced modeling features

Supporting Python Libraries (Recommended):

- pandas for data manipulation
- NumPy for numerical operations
- matplotlib (or similar) for visualization

Documentation Best Practices:

- Create a data dictionary for all variables and parameters
- Document all assumptions with clear justifications
- Include references to academic literature or industry practices for modeling choices
- Provide clear explanations of mathematical notation and modeling logic
- Use Markdown cells in Google Colab to explain:
    - the problem formulation
    - modeling decisions
    - interpretation of results
- Discuss limitations of your approach and potential improvements

9 Resources

The following resources are available to support your project:

- Course textbook (Introduction to Operations Research by Frederick S. Hillier and Gerald J. Lieberman) and lecture slides
- Colab Notebooks with gurobipy code for optimization problems

10 Tips for Success

Success Strategies:

- Start early and plan your project timeline carefully
- Divide tasks among team members based on individual strengths and ensure clear ownership of modeling, analysis, and presentation components
- Build incrementally - start with simple models (LP) and add complexity (Network, MIP, Multi-objective, NLP)
- Document your process and decisions along the way, including assumptions and modeling choices in your Technical Report and Google Colab Notebook
- Test your models with multiple scenarios and client profiles or equivalent operational constraints
- Balance theoretical rigor with practical implementation, ensuring that your model supports decision-making and business insights
- Focus on creating business value through your optimization approach, not only technical correctness
- Prepare for implementation challenges and have contingency plans
- Schedule regular team check-ins to track progress and resolve issues

Effective Project Management:

Planning Phase:

- Create a work breakdown structure aligned with the required deliverables (Technical Report, Presentation, Implementation Package, User Guide)
- Allocate tasks based on team member strengths, ensuring each member contributes to modeling, analysis, and communication
- Set intermediate milestones and deadlines (e.g., formulation, MIP model, results, presentation)
- Schedule regular team check-ins

Development Strategy:

- Start with a simple model and add complexity incrementally (LP → Network → MIP → Multi-objective → NLP)
- Verify each component before integration
- Develop your solution in a Google Colab Notebook using Python and gurobipy, integrating code, results, and explanations
- Maintain version control of your models and code
- Document changes and improvements systematically

Quality Assurance:

- Implement validation checks in your model
- Test with extreme scenarios to identify limitations
- Compare results against simple benchmarks
- Conduct sensitivity analysis on key parameters
- Ensure your results are clearly interpreted and connected to managerial insights and decision recommendations

Common Pitfalls to Avoid:

- Overly complex models that are difficult to explain or implement
- Ignoring data quality issues in the initial stages
- Focusing too much on optimization theory rather than business insights and decision support
- Not allowing sufficient time for model refinement, scenario analysis, and presentation preparation
- Overlooking real-world operational constraints (e.g., client case studies or equivalent)
- Creating visualizations that are technically accurate but difficult to interpret
- Waiting too long to integrate different components of the project
- Underestimating the time needed for testing, refinement, and documentation in the Technical Report and Colab Notebook

11 Important Notes

- This project is designed to apply all the optimization techniques covered in our course.
- While logistics knowledge is helpful, the focus is on applying optimization techniques to the problem.

- **You may make reasonable assumptions when necessary, but document them clearly**.
- Consult with the instructor if you need clarification on any aspect of the project.

12 Team Formation

Team Composition Requirements

- Team size: 4 students per team (except of one team with less than 4 students). There will be 10 teams.
- Teams must be formed and registered by: May 1, 2026
- Registration process: Submit team names and member information via Canvas

12.1 Team Formation Process

Instructions for Student Self-Enrollment in Groups

To sign up for your group, please follow these steps:

1. Open "People": Click the People link in the left-hand navigation menu of our Camino course.
2. Select the Tab: Click the Groups tab at the top of the page.
3. Find Your Group Set: Look for the group category— "Capstone Project Groups"
4. Join an Existing Group: Scroll through the list and click the + Join link on the right side of the group you wish to enter.
5. Note: If a group is full, the "Join" link will disappear.
6. OR Create a New Group: If you are starting a new team, click the + Group button at the top right of the list.
7. Give your group a name and click Save. You will be automatically added to that new group.
8. Verify: Once you have joined or created a group, look for a small checkmark or your name listed under that group to confirm you are successfully enrolled

Step 1: Self-selection period (April 21 - April 28)

Students are encouraged to form their own teams based on complementary skill sets.

Step 2: Matchmaking period (April 29 - April 30)

Students who haven’t found a team by April 28 will be automatically matched.

Step 3: Final registration (May 1)

All teams must be officially registered via Canvas by 11:59 PM.

Important Notes

- Teams will remain the same for the entire project duration
- Any team changes after May 1 require instructor approval and written justification
- Individual contribution assessment will consider peer evaluations

Good luck with your project!

This sustainable urban logistics optimization project offers an excellent opportunity to apply sophisticated optimization techniques to a complex real-world problem with significant environmental and economic impacts.

CAPSTONE CHECKLIST: HOW TO GET A 4 (EXCELLENT)

A correct model gets you a 3. A realistic, decision-ready model gets you a 4.

1. Problem & Data (Clear and Complete)

- Clearly define the decision problem (what decision are you optimizing?)
- Identify objectives: cost, emissions, service level
- Use and explain all relevant datasets
- Document assumptions and data limitations

2. Mathematical Modeling (Required Components)

Core Models

- LP model (baseline)
- Network flow formulation
- MIP model (facility decisions, assignments, logic constraints)

Advanced Components

- Multi-objective optimization (choose ONE method and justify it)
- Nonlinear component (or valid approximation)

3. Operational Realism (REQUIRED FOR A 4)

You must include ONE of the following:

A client case study (C103, C101, C108, or C115), correctly modeled OR Equivalent real-world operational constraints

For full credit:

- Clearly explain the business meaning
- Show correct mathematical formulation
- Justify any assumptions or approximations

4. Solution & Analysis

Compute and present the optimal solution

- Show facility decisions, assignments, and vehicle use
- Perform sensitivity or scenario analysis (≥ 2 parameters)
- Analyze trade-offs (cost vs emissions vs service)

5. Managerial Insights (CRITICAL)

- Translate results into business decisions
- Explain why the solution makes sense
- Provide clear recommendations
- Quantify impact (e.g., cost increase vs service improvement)

6. Technical Implementation (Required Stack)

- Use Google Colab Notebook
- Use Python + gurobipy
- Organize code clearly (sections, comments, structure)
- Use Markdown to explain:
    - model
    - assumptions
    - results

7. Technical Report (Professional Quality)

- Clear structure (problem → model → results → insights)
- Well-defined variables, parameters, constraints
- Clean mathematical notation
- Discussion of limitations and improvements

8. Presentation (Business-Focused)

- Clearly explain the decision problem and solution
- Show key results visually (maps, charts, tables)
- Highlight trade-offs and insights
- Explain impact of operational constraints
- Deliver clear, actionable recommendations

9. Individual Contribution

- You can clearly explain:
    - your part of the model
    - key constraints
    - main insights
- You are prepared for individual Q&A

COMMON REASONS YOU MAY NOT GET A 4

- No client case (or no real-world constraints)
- Only one model (no MIP or no multi-objective)

- Weak or missing analysis
- No business insights (only numbers)
- Poor explanation of the model
- Disorganized Colab Notebook